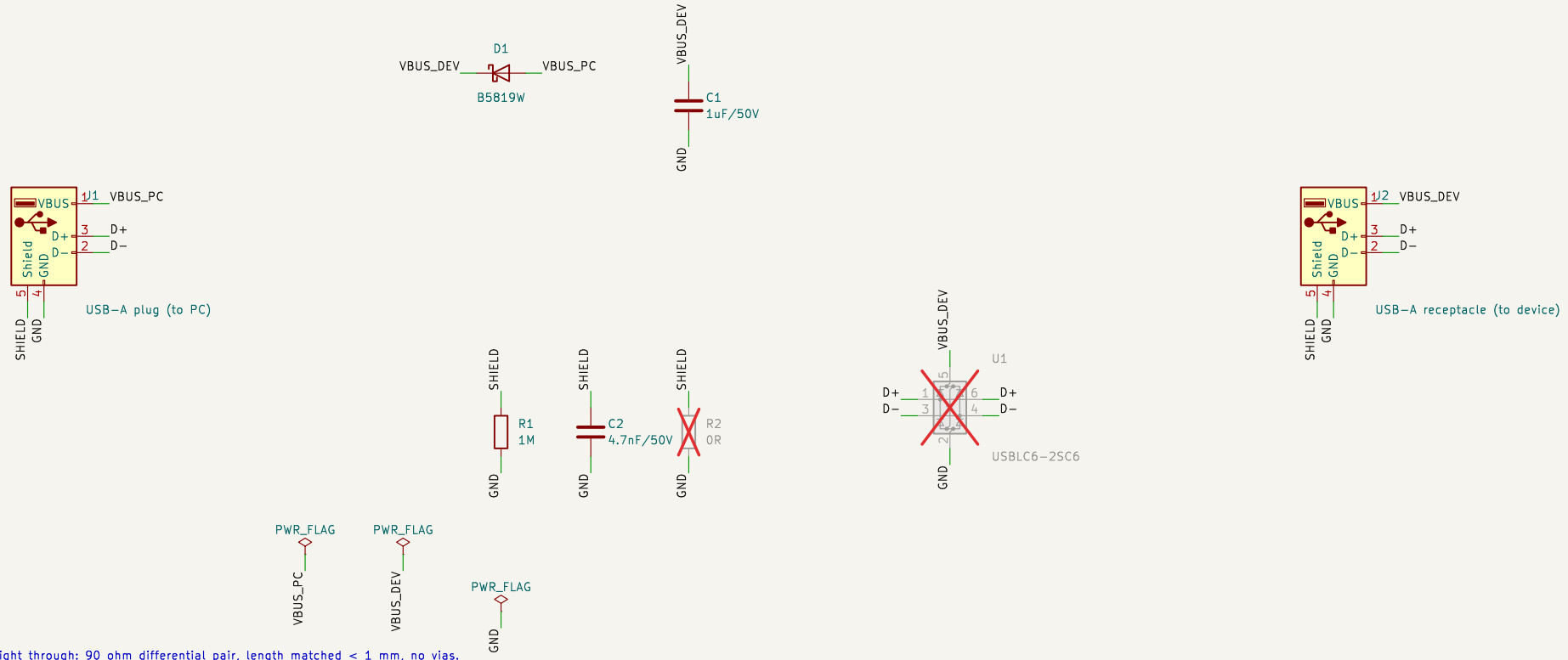
USB-A inline adapter: PC (J1 plug) -> DEV (J2 receptacle), VBUS back-feed blocked by D1



D+/D− pass straight through: 90 ohm differential pair, length matched < 1 mm, no vias.
D1: PC -> DEV only. Device side (main board 5 V, 4.75–5.1 V) cannot back-feed the PC VBUS.
U1 (ESD) and R2 (shield–GND 0R) are not fitted by default.

OpenCruise		
Sheet: /		
File: usbblock.kicad_sch		
Title: OpenCruise USB VBUS-Block adapter		
Size: A4	Date: 2026-11-09	**Rev: 1.0**
KiCad E.D.A. 9.0.9		Id: 1/1